CASE 5
X-LINKED SEVERE COMBINED IMMUNODEFICIENCY

The maturation of T lymphocytes.

Without T cells, life cannot be sustained. In Case 1 we learned that an absence of B cells was compatible with a normal life as long as infusions of immunoglobulin G were maintained. When children are born without T cells, they seem normal for the first few weeks or months. Then they begin to acquire infections, often with opportunistic pathogens, and in the absence of appropriate treatment they die while still in infancy. An absence of functional T cells causes severe combined immunodeficiency (SCID). It is called severe because it is fatal, and called combined because, in humans, B cells cannot function without help from T cells, so that even if the B cells are not directly affected by the defect, both humoral and cell-mediated immunity are lost. Unlike X-linked agammaglobulinemia, which results from a monogenic defect, SCID comprises multiple disorders that can result from any one of several different genetic defects. The incidence of SCID is three times greater in males than in females, and this male:female ratio of 3:1 is due to the fact that the most common form of SCID is X-linked. In the United States, about 25% of individuals with SCID have the X-linked form of the disease.

T-cell precursors migrate to the thymus to mature, at first from the yolk sac of the embryo, and subsequently from the fetal liver and bone marrow (Fig. 5.1). The rudimentary thymus is an epithelial anlage derived from the third and fourth pharyngeal pouches. By 8 weeks of human gestation, the invasion of precursor T cells, and of dendritic cells and macrophages, has transformed the gland into a central

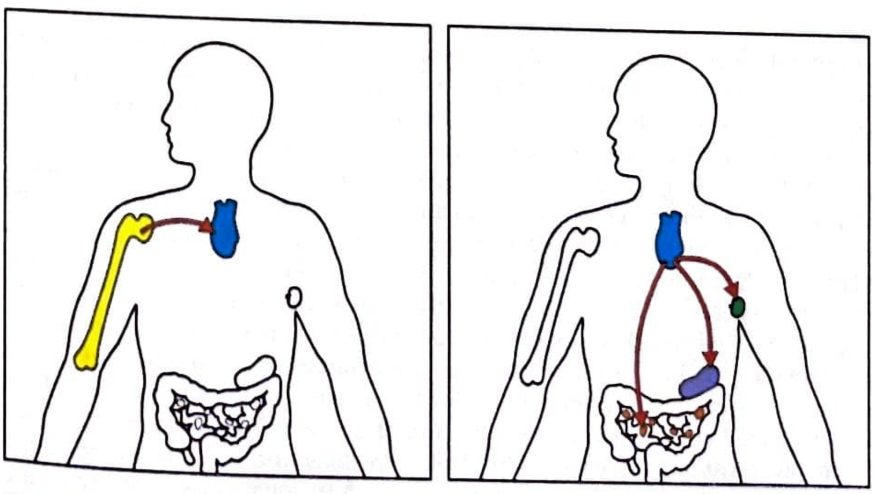
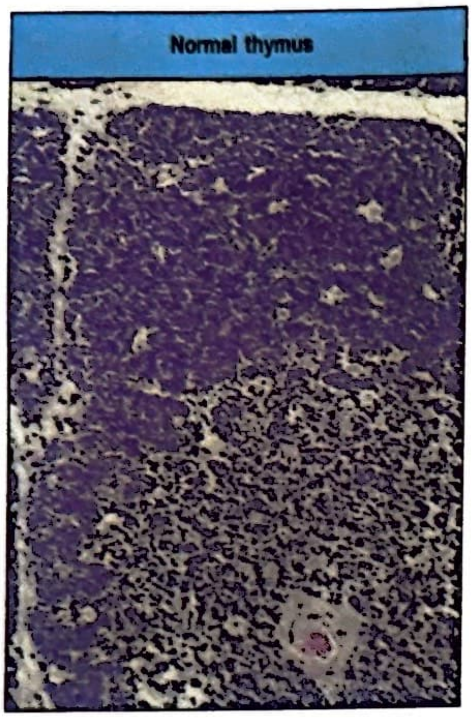
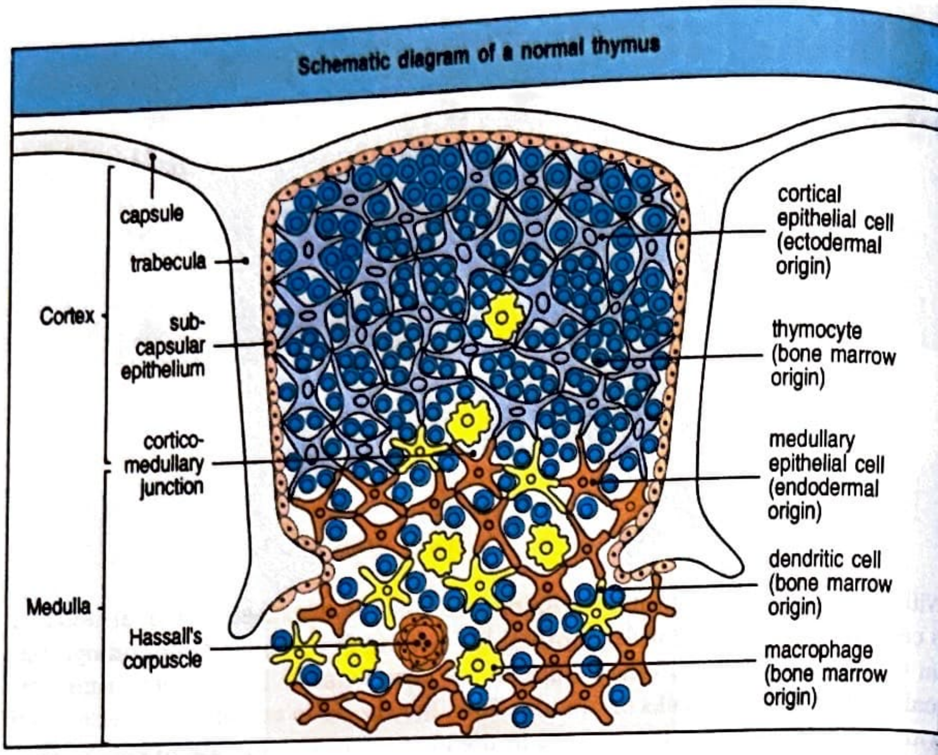


Fig. 5.1 T-cell precursors migrate to the thymus to mature. T cells derive from bone marrow stem cells, whose progeny migrate from the bone marrow to the thymus (left panel), where the development of the T cell takes place. Mature T cells leave the thymus and recirculate from the bloodstream through peripheral lymphoid tissues (right panel) such as lymph nodes, spleen, or Peyer's patches, where they may encounter antigen.

TOPICS BEARING ON THIS CASE:

- The development of T cells in the thymus
- Signal transduction through the T-cell receptor
- Interleukin-2 receptor
- Defects in T-cell function result in severe combined immunodeficiency
- Testing T-cell responses with polyclonal mitogens





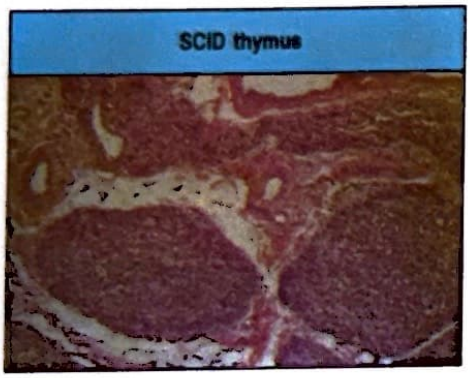


Fig. 5.2 **The cellular organization of the thymus.** The thymus, which lies in the midline of the body above the heart, is made up from several lobules, each of which contains discrete cortical (outer) and medullary (central) regions. The cortex consists of immature thymocytes, i.e., cells of the T lymphocyte lineage that are developing in the thymus (dark blue), branched cortical epithelial cells (pale blue), with which the immature cortical thymocytes are closely associated, and scattered macrophages (yellow) involved in clearing apoptotic thymocytes. The medulla consists of mature thymocytes (dark blue) and medullary epithelial cells (orange), along with macrophages (yellow) and dendritic cells (yellow) of bone marrow origin. Hassall's corpuscles found in the human thymus are probably also sites of cell destruction. The thymocytes in the outer cortical cell layer are proliferating immature cells; the deeper cortical thymocytes are mainly cells undergoing thymic selection. The upper photograph shows the equivalent section of a human thymus, stained with hematoxylin and eosin. The lower photograph shows a SCID thymus. The thymus is small with loss of lymphocytes and no cortico-medullary differentiation. Hassall's corpuscles are absent. Photographs courtesy of C.J. Howe (upper) and A. Perez-Atayde (lower).

lymphoid organ (Fig. 5.2). T-cell precursors undergo rapid maturation in the thymus gland (Fig. 5.3), which becomes the site of the greatest mitotic activity in the developing fetus. By 15 weeks of gestation, mature T cells start to emigrate from the thymus to the peripheral lymphoid organs. In all common forms of SCID the thymus fails to become a central lymphoid organ. A small and dysplastic thymus, as revealed by biopsy, has been used in the past to confirm SCID. Currently, newborn screening for SCID is possible by enumerating T cell receptor excision circles (TRECs), a by-product of rearrangements at the T cell receptor α locus that occur during T cell development (Fig. 5.4). TRECs are maintained within the developing thymocytes, and their levels are progressively diluted as T cells proliferate. Enumeration of TREC levels by quantitative polymerase chain reaction (qPCR) on genomic DNA extracted from a punch biopsy of a newborn screening dried blood spot collected at birth allows identification of newborns with severe T cell lymphopenia, including those with SCID. Definitive diagnosis of SCID is based on the enumeration, phenotypic characterization, and functional analysis of circulating lymphocytes. Furthermore, now that the various gene defects that underlie SCID are better understood, mutation analysis is used to provide definitive diagnosis.

Defects that result in SCID are classified into four general categories. One comprises those defects that impair lymphocyte survival; the prototype of this class

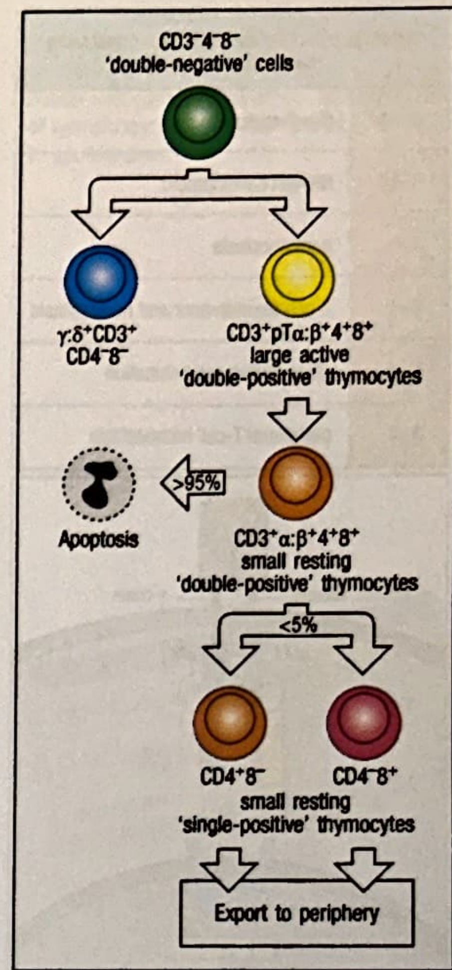


Fig. 5.3 Changes in cell-surface molecules allow thymocyte populations at different stages of maturation to be distinguished. The most important cell-surface molecules for identifying thymocyte subpopulations have been CD4, CD8, and T-cell receptor complex molecules (CD3, and α and β chains). The earliest cell population in the thymus does not express any of these. Because these cells do not express CD4 or CD8, they are called 'double-negative' thymocytes. (The γ:δ T cells found in the thymus also lack CD4 or CD8, but these are a minor population.) Maturation of α:β T cells occurs through stages in which both CD4 and CD8 are expressed by the same cell, along with the pre-T-cell receptor (pTα:β) and, later, low levels of the T-cell receptor (α:β) itself. These α:β cells are known as 'double-positive' thymocytes. Most thymocytes (about 97%) die within the thymus after becoming small double-positive cells. Those whose receptors bind self MHC molecules lose the expression of either CD4 or CD8 and increase the level of expression of the T-cell receptor. The outcome of this process is the 'single-positive' thymocytes, which, after maturation, are exported from the thymus as mature single-positive T cells.

is adenosine deaminase (ADA) deficiency (see Case 6), in which adenosine metabolites that are toxic to T cells and B cells accumulate. Increased apoptosis of T-lymphocyte progenitors and of myeloid precursors is also observed in reticular dysgenesis, which is characterized by extreme lymphopenia, absence of neutrophils, and sensorineural deafness. This disease is due to mutations of the adenylate kinase 2 gene, which regulates levels of ADP (a substrate for ATP production) in the mitochondrial intermembrane space.

A second category of SCID consists of defects in cytokine-mediated signals for lymphocyte maturation and proliferation, and includes the X-linked form of SCID that is the subject of this case. The *IL2RG* gene responsible for the X-linked form of SCID was mapped to the long arm of the X chromosome at Xq11 and then cloned. It encodes the common γ chain (γ_c, CD132) shared by the interleukin-2 receptor (IL-2R) and by other cytokine receptors (IL-4R, IL-7R, IL-9R, IL-15R, and IL-21R) (Fig. 5.5). In all of these cytokine receptors, the γ_c chain associates with the intracellular tyrosine kinase JAK3, encoded by an autosomal gene. JAK3 is essential for intracellular signaling mediated by all γ_c-containing cytokine receptors upon binding of the respective cytokine.

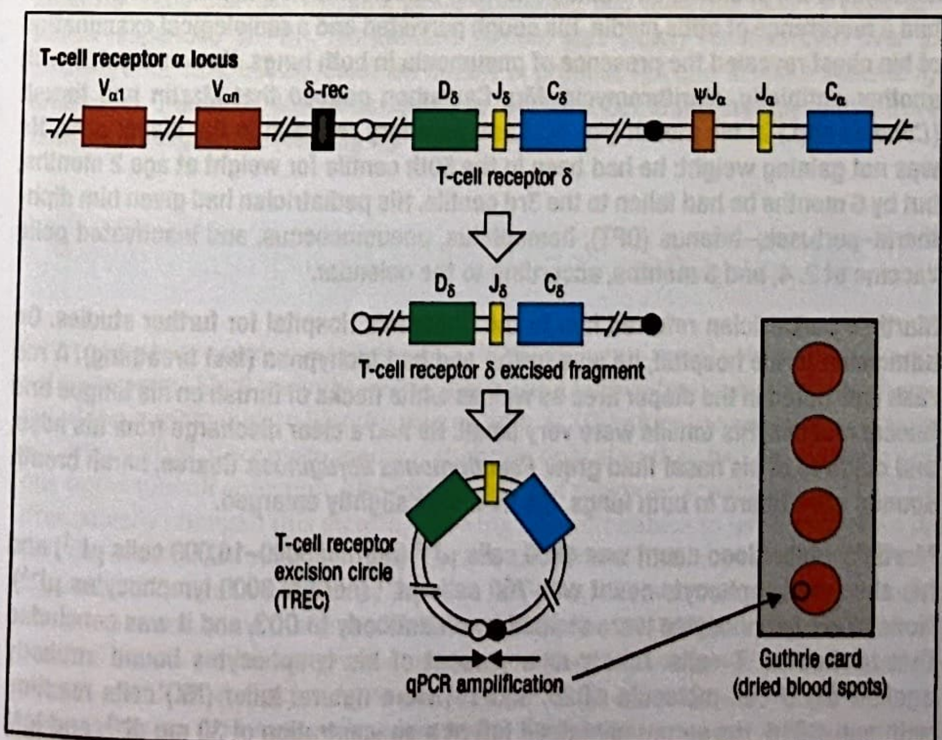


Fig. 5.4 Enumeration of T cell receptor excision circles (TRECs) permits diagnosis of SCID at birth. During thymocyte development, rearrangement at the T cell receptor α (*TCRA*) locus involves deletion of TCR δ (*TCRD*) gene elements, which are circularized, forming T cell receptor excision circles (TRECs). TRECs are retained intracellularly during differentiation to fully mature T lymphocytes, but are progressively diluted as a result of T cell proliferation. Enumeration of TRECs in genomic DNA extracted from a punch biopsy of dried blood spot collected at birth is a measure of thymopoiesis. Infants with SCID have undetectable or very low levels of TRECs.

Cytokines that act on receptors containing the common γ chain	
IL-21	B-cell maturation
IL-15	NK-cell development
IL-9	hematopoiesis
IL-7	T-cell development and homeostasis
IL-4	class-switch recombination
IL-2	peripheral T-cell homeostasis

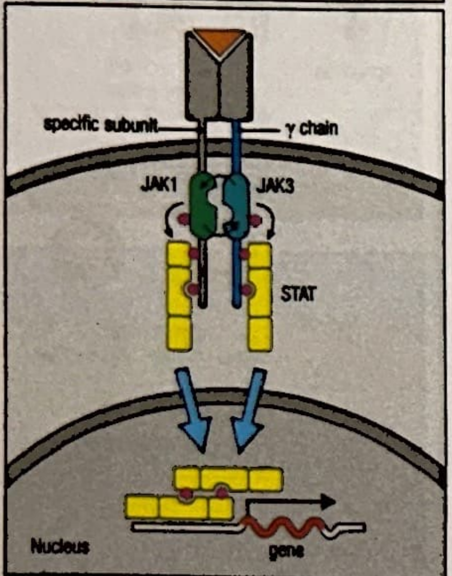


Fig. 5.5 The γ_c chain is a component of multiple cytokine receptors with distinct roles in immune function. The γ_c chain is shared by cytokine receptors for IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21, which have distinct roles in T-cell, B-cell, and NK-cell development and function. Binding of ligand (*orange triangle*) to the cytokine receptor initiates a JAK tyrosine kinase signaling cascade, which phosphorylates a number of STAT (signal transducers and activators of transcription) proteins. The STAT proteins translocate to the nucleus, bind to various genes, and activate their expression.

A third category of SCID consists of defects in the machinery for somatic gene rearrangement that assembles the immunoglobulin and T-cell receptor (TCR) genes during lymphocyte development, the so-called V(D)J recombination process. This category can be divided into lymphocyte-specific defects, namely defects in the *RAG* genes that encode the lymphocyte-specific recombinase (see Case 7), and defects in ubiquitously expressed DNA repair genes that are also involved in this recombination.

Finally, another group of SCID is due to mutations in genes that encode the CD3 molecules that participate in the formation of the CD3:TCR complex. This complex allows signaling through the pre-T-cell receptor.

Other forms of combined immunodeficiency are due to defects that affect later stages in T-cell development. In these cases, there is a residual presence of circulating T lymphocytes, but their function is impaired.

The case of Martin Causubon: without T cells, life cannot be sustained.

Mr and Mrs Causubon had a normal daughter 3 years after they were married. Two years later they had a son and named him Martin. He weighed 3.5 kg at birth and seemed to be perfectly normal. At 3 months of age, Martin developed a runny nose and a persistent dry cough. One month later he had a middle ear infection (otitis media) and his pediatrician treated him with amoxicillin. At 5 months of age Martin had a recurrence of otitis media. His cough persisted and a radiological examination of his chest revealed the presence of pneumonia in both lungs. He was treated with another antibiotic, clarithromycin. Mrs Causubon noticed that Martin had thrush (*Candida* spp.) in his mouth (Fig. 5.6) and an angry red rash in the diaper area. He was not gaining weight; he had been in the 50th centile for weight at age 2 months, but by 6 months he had fallen to the 3rd centile. His pediatrician had given him diphtheria–pertussis–tetanus (DPT), hemophilus, pneumococcus, and inactivated polio vaccine at 2, 4, and 6 months, according to the calendar.

Normal B cells but no T cells.

Martin's pediatrician referred him to the Children's Hospital for further studies. On admission to the hospital, he was fretful and had tachypnea (fast breathing). A red rash was noted in the diaper area as well as white flecks of thrush on his tongue and buccal mucosa. His tonsils were very small. He had a clear discharge from his nose, and cultures of his nasal fluid grew *Pseudomonas aeruginosa*. Coarse, harsh breath sounds were heard in both lungs. His liver was slightly enlarged.

Martin's white blood count was 4800 cells μl^{-1} (normal 5000–10,000 cells μl^{-1}) and his absolute lymphocyte count was 760 cells μl^{-1} (normal 3000 lymphocytes μl^{-1}). None of his lymphocytes were stained by an antibody to CD3, and it was concluded that he had no T cells. Ninety-nine percent of his lymphocytes bound antibody against the B-cell molecule CD20, and 1% were natural killer (NK) cells reacting with anti-CD16. His serum contained IgG at a concentration of 30 mg dl^{-1}, and IgM

Fig. 5.6 An infant with SCID suffering from *Candida albicans* in the mouth.

Mitogen	Abbreviation	Source	Responding cells
Phytohemagglutinin	PHA	*Phaseolus vulgaris* (red kidney beans)	T cells (human)
Concanavalin A	ConA	*Canavalia ensiformis* (jack bean)	T cells
Pokeweed mitogen	PWM	*Phytolacca americana* (pokeweed)	T and B cells
Lipopolysaccharide	LPS	*Escherichia coli*	B cells (mouse)

Fig. 5.7 Polyclonal mitogens, many of plant origin, stimulate lymphocyte proliferation in tissue culture. Many of these mitogens are used to test the ability of lymphocytes in human peripheral blood to proliferate.

at 12 mg dl^{-1}. Serum IgA were undetectable. His blood mononuclear cells were completely unresponsive to phytohemagglutinin (PHA), concanavalin A, and pokeweed mitogen (Fig. 5.7), as well as to specific antigens to which he had been previously exposed by immunization or infection—tetanus and diphtheria toxoids, and to *Candida* antigen. His red cells contained normal amounts of adenosine deaminase and purine nucleoside phosphorylase. Cultures of sputum for bacteria and viruses revealed the abundant presence of respiratory syncytial virus (RSV).

No HLA-identical sibling.

At this point a blood sample was obtained from Martin's mother to examine her T cells for random inactivation of the X chromosome (a diagnostic test explained in the answer to Question 1 in Case 1). It was found that her T cells exhibited complete nonrandom X-chromosome inactivation. Search for mutations in the *IL2RG* gene revealed deletion of a single A nucleotide in exon 2, leading to a frameshift. HLA typing showed that Martin's sister had no matching HLA alleles. His parents, as expected, each shared one HLA haplotype with Martin.

Mother agrees to donate bone marrow.

Martin was treated with intravenous gamma globulin at a dose of 0.5 g kg^{-1} body weight every 3 weeks, and his serum IgG level was maintained at 600 mg dl^{-1} by subsequent IgG infusions. He was given aerosolized ribavirin to control his RSV infection and trimethoprim-sulfamethoxazole for prophylaxis against *Pneumocystis jirovecii*. Without any chemotherapy, Martin was given 5×10^6 kg^{-1} $CD34^+$ cells that had been purified from his mother's bone marrow. Three months after receiving the maternal bone marrow, Martin's blood contained 600 maternal $CD3^+$ T cells μl^{-1}, which responded to PHA. His immune system was slowly reconstituted over the ensuing 3 months, but he remained unable to produce IgA and to make specific IgG antibodies. Therefore, Martin continues to require substitution therapy with intravenous immunoglobulin (400 mg kg^{-1} every 3 weeks).

Engraftment is successful.

Severe combined immunodeficiency (SCID).

Severe combined immunodeficiency, or SCID, presents the physician with a medical emergency. Until recently, unless there was a known family history, which provided the opportunity to identify infants with SCID before the onset of infections, children with SCID came to medical attention only after they had developed a serious opportunistic infection. The introduction of newborn screening for SCID has dramatically changed this picture, allowing affected babies to be identified in the first days of life, followed by immediate intervention. Although newborn screening for SCID is available in most states in the USA, it is not routinely performed in many other countries in the world. In most cases of SCID, the first symptoms are those of thrush in the mouth and diaper area. A persistent cough usually betrays infection with *Pneumocystis jirovecii*. Interstitial pneumonia may also be due to viral infections, and especially adenovirus, RSV, parainfluenza virus type 3, and cytomegalovirus. The third most common symptom of SCID is intractable diarrhea, often due to enteropathic coliform bacilli or viruses. Because infants with SCID die rapidly from infections, measures must be taken quickly to prevent infections and to reconstitute their immune system. Preventive measures include use of prophylactic trimethoprim-sulfamethoxazole starting at 8 weeks of life, regular use of intravenous immunoglobulins, and hygiene measures, including avoidance of contact with individuals with active infections. Administration of live viral vaccines (including the rotavirus vaccine, which is typically administered at 2 months of life in healthy infants) must be avoided, as these vaccines can cause serious and even fatal infections in SCID babies. The only definitive cure is provided by reconstituting the immune system, typically with hematopoietic cell transplantation (HCT).

As previously noted, SCID has many known genetic causes. Among autosomal recessive forms of SCID, deficiency of adenosine deaminase (ADA) (see Case 6) or mutations in the *RAG* genes, which initiate V(D)J recombination (see Case 7), are more common. In most patients with ADA deficiency, and in patients with severe *RAG* mutations, both T and B lymphocytes are absent. In contrast, patients with X-linked SCID lack circulating T lymphocytes but have a normal number of B cells. Natural killer lymphocytes (NK cells) are also absent.

Other cases of autosomal recessive SCID resemble the phenotype of X-linked SCID and have been ascribed to defects in the protein JAK3, which transduces the signal from γ_c-containing cytokine receptors. Therefore, patients with JAK3 deficiency also lack T and NK cells and have a normal number of B lymphocytes, similar to the situation observed in patients with X-linked SCID.

Mice lacking IL-2 do not develop SCID, which argues against an important function for IL-2 signaling in thymocyte development. The discovery that mutations in the IL-2 receptor γ chain (IL-2Rγ) caused X-linked SCID in humans seemed inconsistent with this finding in mice. This led to a search for the γ chain in other interleukin receptors that might be more important for lymphocyte development. It was found that this chain also forms part of the receptors for IL-4, IL-7, IL-9, IL-15, and IL-21, and it was renamed the common gamma (γ_c) chain. Among the cytokines that bind to γ_c-containing cytokine receptors, IL-7 has a critical role in T-cell development. In addition, IL-7 is essential for B-cell development in mice, but not in humans. *Il7*$^{-/-}$ and *Il7r*$^{-/-}$ mice lack both T and B lymphocytes; in contrast, patients with SCID due to mutations of the *IL7R* gene lack circulating T lymphocytes but have a normal number of B and NK cells. Finally, the absence of NK cells in patients with X-linked SCID and with JAK3 deficiency is due to impaired signaling through the IL-15R, as shown by impaired development of NK cells in *Il15*$^{-/-}$ and *Il15r*$^{-/-}$ mice.

Mutations of the *RAG* genes and of other genes involved in V(D)J recombination impair rearrangement of the T-cell receptor and immunoglobulin genes, and abrogate the development of both T and B lymphocytes, whereas the development of NK lymphocytes is not affected. In contrast, patients with SCID due to mutations in the genes that encode the CD3δ, CD3ε, and CD3ζ chains of the CD3 complex have selective T-cell deficiency.

Later defects in T-cell development include mutations in proteins that transduce signals from the T-cell receptor, and therefore are associated with impaired function of peripheral T cells. These defects are discussed in Cases 54 and 55.

SCID is fatal, unless reconstitution of T-cell immunity is achieved. Hematopoietic cell transplantation (HCT) is the treatment of choice in most cases of SCID. When an HLA-matched family donor is available, cure can be achieved in more than 90% of infants with SCID. In other cases, transplantation from a partially matched (haploidentical) family donor or from a matched unrelated donor can be used. If transplantation is performed in the first 3.5 months of life or prior to the development of infections, survival is excellent (around 80-90%), regardless of the type of donor (HLA-matched sibling, haploidentical parent, or matched unrelated donor). However, results are less satisfactory if the patient has active infections at the time of transplantation. In some cases, HCT can also correct antibody deficiency in patients with SCID; in the remaining cases, immunoglobulin replacement therapy remains necessary. Gene therapy (using retroviral or lentiviral vectors that allow expression of the γ_c chain) has also been successful in some patients with X-linked SCID; however, leukemic proliferation has been observed in some of the patients treated with earlier versions of retroviral vectors, as the result of activation of oncogenes at the site of retroviral integration. Safer vectors have been developed recently that allow immune reconstitution without development of leukemia; however, long-term follow up studies are needed to confirm safety.

Questions.

1 *Martin was suspected to have X-linked SCID because he had a severe deficiency of T and NK lymphocytes, with a normal number of B lymphocytes. Even before the mutation in the IL2RG gene was identified, which test proved that he had X-linked SCID?*

2 *What is known about the normal functions of the receptors of which the γ_c chain forms a part, and how might these account for the phenotype of X-linked SCID?*

3 *Why was it necessary to treat the maternal bone marrow and purify CD34$^+$ cells? What other strategies could have been used?*

4 *Frequently infants with SCID get very ill with Pneumocystis jirovecii pneumonia after a successful transplant. For this reason they are treated prophylactically with antibiotics to get rid of any P. jirovecii organisms that may be present in their lungs. Why is this a wise therapeutic maneuver, and how do you explain the worsening of pneumonia after a transplant?*

5 *We have already discussed the risks associated with giving live poliovirus vaccine to immunodeficient infants in the case of X-linked agammaglobulinemia (see Case 1). Martin received a killed (inactivated) polio vaccine. Had he received a live attenuated polio vaccine instead, he could have developed poliomyelitis. Fortunately, Martin escaped being given any live vaccines before he was diagnosed. In many countries (but not the United States) infants are universally given Bacillus Calmette–Guérin (BCG), an attenuated form of the tuberculosis bacillus, which provides partial protection against tuberculosis infection. BCG incites a cell-mediated immune response and after receiving it infants become tuberculin-positive, which means they show a delayed-type hypersensitivity response to a skin-prick with minute quantities of tuberculin. In the United States, the tuberculin test is considered so diagnostically valuable for the detection of new tuberculosis infections that BCG is not given. What do you think happens to infants with SCID who are given BCG?*

6 *Before smallpox was eradicated in the world, vaccinia virus was routinely administered to all children. What happened to infants with SCID who were vaccinated with vaccinia virus?*

7 *Although transplantation of maternal bone marrow cells resulted in successful T-cell reconstitution, Martin remained unable to produce antibodies. This failure of*

B-cell reconstitution is often observed in patients with X-linked SCID or with JAK deficiency, even when the transplant is performed from an HLA-identical donor. What could be the underlying mechanism?

8 *What would be another strategy for reconstituting immune function in Martin?*

CASE 6
ADENOSINE DEAMINASE DEFICIENCY

The purine degradation pathway and lymphocytes.

In Case 5 we learned about severe combined immunodeficiency (SCID) and examined the most common form—that due to an X-linked mutation in the gamma chain (γ_c) common to several interleukin receptors. In that type of SCID, T cells are virtually absent, whereas B cells are present in normal numbers although they are not functional. Further examination of this phenotype also reveals an absence of natural killer (NK) cells. Thus, X-linked SCID is classified as $T^-B^+NK^-$ SCID.

SCID patients with an almost complete absence of both B cells and T cells are also encountered; their phenotype is T^-B^-. This phenotype is exclusively associated with SCID transmitted as an autosomal recessive condition. Some forms of T^-B^- SCID are due to defects of V(D)J recombination; these disorders have a normal number of NK lymphocytes ($T^-B^-NK^+$ SCID). However, in other cases of T^-B^- SCID, the number of NK lymphocytes is severely reduced ($T^-B^-NK^-$ SCID). The most common genetic defect encountered in these patients is mutation in the gene encoding the purine-degradation enzyme adenosine deaminase (ADA). ADA is a ubiquitous housekeeping enzyme found in all mammalian cells; it converts the purine nucleosides adenosine and deoxyadenosine to inosine and deoxyinosine respectively, and hence to waste products that are excreted (Fig. 6.1). In its absence, cells can accumulate excessive amounts of phosphorylated adenosine and deoxyadenosine metabolites, which are toxic when present in excessive amounts. Lymphocytes are particularly susceptible to the toxic effects of these metabolites, and thus ADA

TOPICS BEARING ON THIS CASE:

- Defects in lymphocyte function result in severe combined immunodeficiency
- Bone marrow transplantation
- Mixed lymphocyte reaction
- Minor histocompatibility antigens
- Graft-versus-host reaction